

4.1.2 Database Management System (DBMS)

ڈی بی ایم ایس وہ سافٹ ویئر ہے جو صارف اور ڈیٹا بیس کے درمیان پل کا کام کرتا ہے۔

CS XI

Chapter 4: Data and Analysis

Lecture 02 of 15

4.1.2 Database Management System (DBMS)

LEARN IT

A **DBMS** is software that lets you create, manage and interact with databases. It sits between the user and the stored files. You never edit raw data files by hand — you talk to the DBMS (Access, MySQL, Oracle, SQLite) and it handles storage, security and consistency.

Advantages you must be able to list:

- **Less redundancy.** The department name is stored once, not in every student row.
- **Consistency.** Change a phone number once; every screen that uses it shows the new number.
- **Security.** Only authorised users see sensitive fields (an admin sees marks; office staff see names).
- **Integrity.** Rules stop duplicate roll numbers and empty primary keys.

WORKED EXAMPLE

In a paper register, Ali's address is written on the admission page, the fee page and the sports page. If he moves house, two of the three pages stay wrong. In a DBMS, `Address` lives in one **Student** table. Every form and report reads that single field.

اردو میں سمجھیں

ڈی بی ایم ایس وہ سافٹ ویئر ہے جو صارف اور ڈیٹا بیس کے درمیان پل کا کام کرتا ہے۔ اس سے تکرار کم ہوتی ہے، ڈیٹا ایک جیسا رہتا ہے، صرف مجاز افراد دیکھ سکتے ہیں، اور غلط اندراج رک جاتی ہے۔

CHECK YOURSELF

1. Name two DBMS programs.
2. Which advantage stops two students sharing the same roll number?

Answers: 1. MS Access, MySQL (Oracle, SQLite also). 2. Integrity (unique / primary-key constraint).

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. A DBMS is software that:

- a. Creates and manages databases
- b. Draws only pie charts
- c. Routes IP
- d. Sorts using bubble only

MCQ 2. Examples of DBMS:

- a. MS Access, MySQL, SQLite, Oracle
- b. Python only
- c. Figma only
- d. Chrome only

MCQ 3. DBMS handles:

- a. Storage, retrieval, security, multi-user access
- b. Only fonts
- c. Only OSI layer 1
- d. Only NAND

MCQ 4. Without a DBMS, programs would:

- a. Each invent their own file format
- b. Run faster always
- c. Need no keys
- d. Need no SQL

MCQ 5. Access is used in XI because:

- a. It is a GUI DBMS for tables/forms/queries/reports
- b. It is a programming language
- c. It is Agile
- d. It is analog

MCQ 6. SQLite in XII is a:

- a. Lightweight DBMS/library
- b. Chart
- c. Gate
- d. Sprint

Extra short questions

1. Define DBMS.
2. Four DBMS functions.
3. DBMS vs database.
4. Name two desktop and two server DBMS.

Extra long question

1. Compare managing 500 students in Excel vs a DBMS (Access). Use data integrity, search, multi-user and reports.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Database Management System: software to define, store, query, protect and share a database.
2. Create tables, insert/update/delete, query, enforce keys/security, backup.
3. Database is the data; DBMS is the software that manages it.
4. Desktop: Access, SQLite. Server: MySQL, Oracle, PostgreSQL.

Long

1. Excel: easy but duplicates, weak keys, clash when two people edit. DBMS: primary keys, queries, forms, reports, controlled access.

Next lecture: 4.2 Database Components (Tables, Records, Fields)